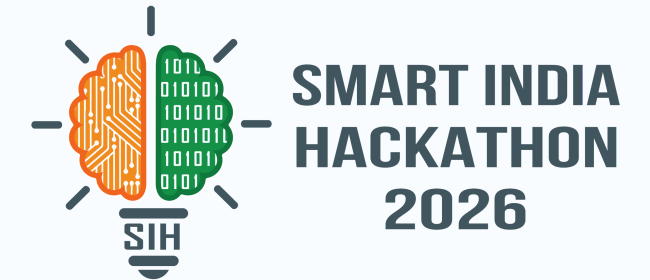


SMART INDIA HACKATHON 2026



Problem Statement ID - SIH26209 Student Innovation

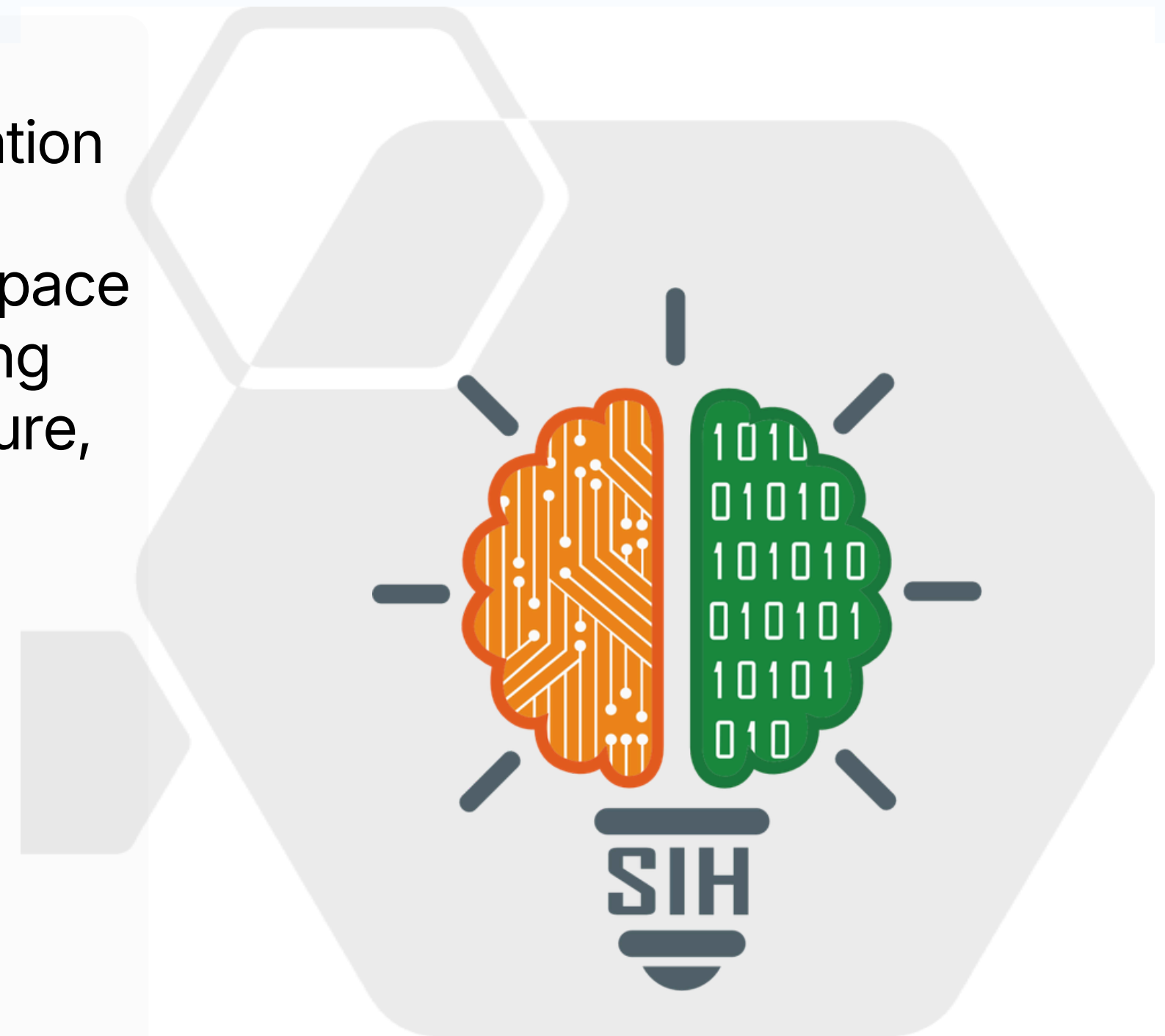
Problem Statement Title - Student Innovation-Space technology refers to the application of engineering principles to the design, development, manufacture, and operation of devices and systems for space travel and exploration.

Theme - Space Technology

PS Category - Software

Team ID - 177054

Team Name - CookieBytes



Nabh-Drishti : Geospatial Disaster Monitoring

PROJECT ABSTRACT:

An adaptive geospatial monitoring framework that intelligently routes heterogeneous satellite observations to specialized expert models for fast, scalable, and accurate detection and tracking of disaster-related changes across optical, SAR, and thermal data.

THE PROBLEM:

- **Orbital Data Glut:** Earth observation constellations (Sentinel, INSAT, Landsat) generate terabytes of imagery every day.. Manual processing creates fatal latency.
- **Sensor Heterogeneity:** Optical, SAR, and thermal radiometers require isolated pipelines, resulting in fragmented ground-segment software.
- **Compute Bottlenecks:** Running dense monolithic models on global 10m-30m raster tiles overloads ground-station servers.



PROPOSED SOLUTION:

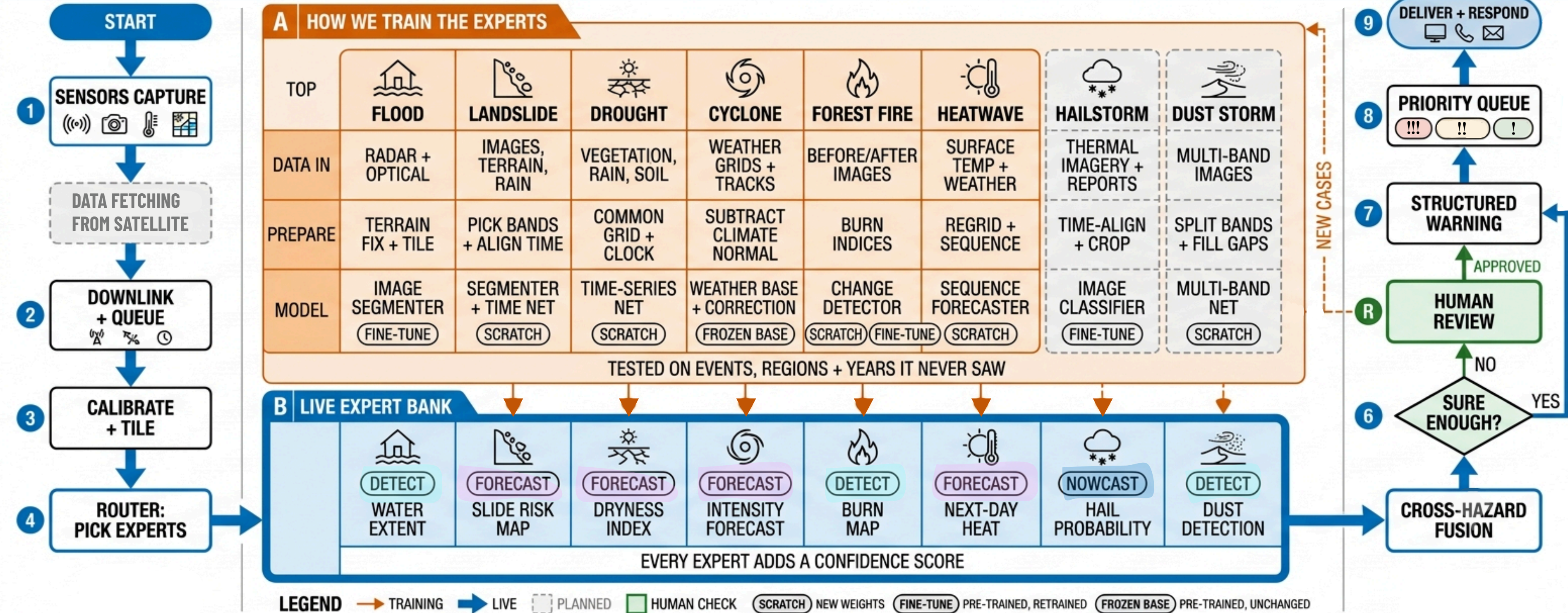
- **Routed Mixture-of-Experts (MoE):** Automatically parses multi-constellation satellite passes, routing specific orbital bands (SAR for surface deformation/water, MWIR/TIR for persistent thermal anomalies) to specialized expert heads.
- **Geospatial Foundation Model Backbones:** Leverages frozen Prithvi-EO and Prithvi-WxC representations, achieving high temporal and spatial transferability with low fine-tuning overhead.

Technical Approach :

Train offline (A), route live (B): confidence-aware pipeline



SMART INDIA
HACKATHON
2026



Technical Feasibility

- Built on openly released pre-trained models (Prithvi-EO, Prithvi-WxC), so no foundation-model pre-training is needed.
- Runs on free, open data: Sentinel-1/2, Landsat, MODIS/VIIRS, CHIRPS, ERA5, plus INSAT (MOSDAC) and Bhuvan for India.
- Public benchmarks exist for the experts: Sen1Floods11 (flood), Landslide4Sense (landslide), D-Fire (fire).
- The router activates only the relevant experts per tile, which keeps ground-station compute low.



Challenges and Mitigation

- Cloud cover and night: SAR and thermal bands complement optical.
- Scarce labels: frozen backbones, light fine-tuning, public benchmarks.
- False alarms: per-expert confidence scores, cross-hazard fusion, and human review before any alert.
- Generalization: tested on events, regions and years never seen in training.



Impact:

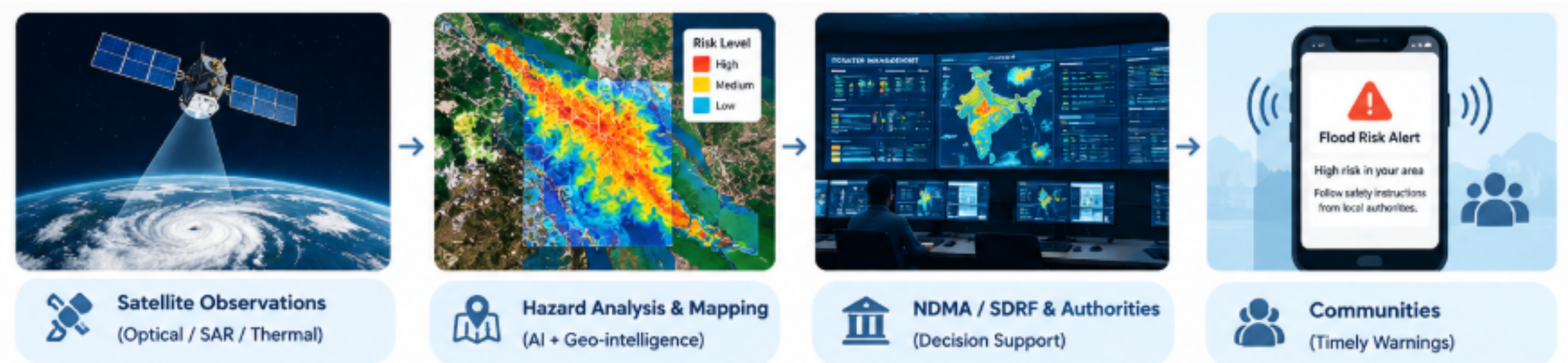
- **Disaster Management Agencies:** Unified satellite intelligence for faster response decisions.
- **Local Administration:** Hazard-specific maps to identify vulnerable areas and prioritize evacuation.
- **Communities:** Clear, location-specific warnings through authorized channels.

Benefits:

- **Social:** Supports timely evacuation and protection of lives and livestock.
- **Economic:** Helps reduce potential losses to infrastructure and agriculture.
- **Environmental:** Enables large-area monitoring of disaster and environmental changes.
- **Operational:** Hazard-aware routing enables scalable and efficient processing.

Why our solution is different?

One router replaces separate per-sensor pipelines, and new hazards plugin as new experts without retraining the whole system (**MODULARITY**).



SOME OF THE PAPER as REFERENCES

- Geospatial foundation models for image analysis: evaluating and enhancing NASA-IBM Prithvi's domain adaptability
<https://arxiv.org/pdf/2409.00489>
- Application of Deep Learning Architectures for Satellite Image Time Series Prediction: A Review (ResearchGate)
<https://www.researchgate.net/publication/356679309>
- Global weather forecasting through deep learning: applications and challenges (ScienceOpen)
<https://www.sciopen.com/article/10.26599/TRCN.2025.9550021>
- Official / Government Data Sources (India):
IMD, Bhuvan (ISRO), GSI, NCS, India-WRIS / CWC, INCOIS, NRSC, FSI
- Synthetic Aperture Radar (SAR) Specialization
[Deep Learning Meets SAR: Concepts, Models, Pitfalls, and Perspectives](#)

Benchmark Datasets & Repositories

- Sen1Floods11: 11-event flood benchmark (Sentinel-1 SAR + hand-labeled masks)
- DeepSARFlood 13-event set: used for ensemble diversity-selection validation
- Landslide4Sense / GVLM-S2: multi-region landslide benchmark (Japan, Nepal, Taiwan, India)
- NASA Global Landslide Catalog
- D-Fire dataset: 21,000 labeled fire/smoke images, YOLO bounding-box format

All our research and collected data: [Miro Whiteboard](#)
Link to Protoype Repo: [GitHub Repo](#)

Global Satellite & Reanalysis Data

- NASA MODIS / VIIRS: NDVI, LST, active-fire detection (FIRMS), Aerosol Optical Depth
- Sentinel-1 (SAR) / Sentinel-2 (optical): ESA Copernicus program
- CHIRPS: satellite-derived rainfall
- ERA5 (Copernicus Climate Data Store): reanalysis: temperature, wind, CAPE, soil moisture
- GRACE / GRACE-FO: groundwater anomaly
- SMAP (NASA): soil moisture
- GLDAS: Global Land Data Assimilation System
- INSAT-3D/3DR (via MOSDAC): geostationary imagery for cyclone/hail
- USGS: ANSS ComCat earthquake catalog, SRTM DEM, Landsat
- IBTrACS / JTWC / HURDAT: cyclone best-track datasets
- GridSat: geostationary IR imagery (cyclone intensity, hail)
- MERRA-2: reanalysis grid used by Prithvi WxC